

RMO– 2019

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Suppose x is a nonzero number such that both x^5 and $20x + \frac{19}{x}$ are rational numbers. Prove that x is a rational number.
2. Let ABC be a triangle with circumference Ω and let G be the centroid of triangle ABC . Extend AG , BG and CG to meet the circle Ω again in A_1 , B_1 and C_1 respectively. Suppose $\angle BAC = \angle A_1B_1C_1$, $\angle ABC = \angle A_1C_1B_1$ and $\angle ACB = \angle B_1A_1C_1$. Prove that ABC and $A_1B_1C_1$ are equilateral triangles.
3. Let a, b, c be positive real numbers such that $a+b+c = 1$. Prove that

$$\frac{a}{a^2 + b^3 + c^3} + \frac{b}{b^2 + c^3 + a^3} + \frac{c}{c^2 + a^3 + b^3} \leq \frac{1}{5abc}$$
4. Considering the following 3×2 array formed by using the numbers 1, 2, 3, 4, 5, 6:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 2 & 5 \\ 3 & 4 \end{pmatrix}.$$
 Observe that all row sum are equal, but the sum of the squares is not the same for each row. Extend the above array to a $3 \times k$ array $(a_{ij})_{3 \times k}$ for a suitable k , adding more columns, using the numbers 7, 8, 9, ..., $3k$ such that

$$\sum_{j=1}^k a_{1j} = \sum_{j=1}^k a_{2j} = \sum_{j=1}^k a_{3j} \text{ and } \sum_{j=1}^k (a_{1j})^2 = \sum_{j=1}^k (a_{2j})^2 = \sum_{j=1}^k (a_{3j})^2.$$
5. In a triangle ABC , let H be the orthocenter, and let D, E, F be the feet of the altitudes from A, B, C to the opposite sides respectively. Let L, M, N be mid points of segments AH, EF, BC respectively. Let X, Y be the feet of altitudes from L, N on to the line DF . Prove that XM is perpendicular to MY .
6. Suppose 91 distinct positive integer greater than 1 are given such that there are at least 456 pairs among them which are relatively prime. Show that one can find four integers a, b, c, d among them such that $\gcd(a, b) = \gcd(b, c) = \gcd(c, d) = \gcd(d, a) = 1$.

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